Exercise 6

a) $X = \overline{A + B}$

b) $X = ABCD$

c) $X = \overline{ABCD}$

d) $X = ABC$

e) $X = A + B + C + D + E$

f) $X = \overline{AB + CD)\, EF + GH)}$

Exercise 13

$X = \overline{ABC} + \overline{A}B\overline{C} + A\overline{BC} + ABC$

Exercise 20

a) $X = \overline{\overline{AB} \cdot \overline{C} \cdot \overline{D})}$

b) $X = \overline{\left(\overline{\overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D})}\right)}$

Exercise 4

Binary Inputs: A = 111 7), B = 101 5)

Sum Output: 1100 12)

Exercise 8

t1: 000

t2: 011

t3: 101

t4: 101

t5: 001

t6: 101

t7: 101

t8: 111

Exercise 14

t1: A = B

t2: A > B

t3: A > B

t4: A > B

t5: A < B

Exercise 16

a) 0101

b) 1010

c) 0011

d) 1100

Exercise 28

$Y = 1$